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Table 3 Changes in initial parameters for water content and SEI to study the effect of different manufacturing and ageing conditions: reference scenario, organic or inorganic SEI, thick or thin SEI, wet or dry electrode

	Scenario						
Parameter	Reference	Thick SEI	Thin SEI	Organic SEI	Inorganic SEI	Wet electrode	Dry electrode
d_{SEI}/nm	50	**75**	**25**	50	50	50	50
ε_{LEDC}/vol%	45	45	45	**90**	**0**	46	45
$\varepsilon_{Li_2CO_3}$/vol%	34	34	33	**8.4**	**39**	30	34
ε_{LiF}/vol%	10.4	10.4	11.4	**1**	**30.4**	9	10.6
ε_{Li_2O}/vol%	10	10	10	**0**	**30**	10	10
ε_{LiOH}/vol%	0.6	0.6	0.6	0.6	0.6	**5**	**0.4**
C_{H_2O}/ppm	260	260	260	260	260	**505**	**168**
C_{HF}/ppm	0	0	0	0	0	**0**	**0**
C_{PF_5}/ppm	992	992	992	992	992	**880**	**1017**
C_{POF_3}/ppm	46	46	46	46	46	**95**	**31**
$C_{HPO_2F_2}$/ppm	1306	1306	1306	1306	1306	**5332**	**536**

3 Results and discussion

The upcoming section evaluates the propagation and characteristics of the thermal runaway during thermal abuse of the given Li-ion battery for the various manufacturing and ageing scenarios. First, sensitivities of ARC measurements to water content, SEI composition and SEI thickness are identified and quantified. This is followed by an in-depth analysis of this behaviour and the underlying causes based on the progression of reactions, their interplay, and their contribution to the temperature evolution during the abuse test. Particular focus is given to the transition from the heat–wait–seek to the exothermal self-heating mode, as this is the crucial point determining a battery's safety range. Then we elucidate the effects of different water contents and initial SEI properties on the state of the battery and occurring reactions and their consequences in an ARC measurement.

3.1 Impact of the SEI state and H_2O on temperature evolution

In Fig. 2, the simulated temperature evolution is displayed for a variation in (a) SEI composition, (b) SEI thickness, and (c) H_2O content. The preheating procedure (1 °C min^{-1}) can be observed up to 40 °C. From then on, the ARC switches to the heat–wait–seek phase and performs 10 °C heating steps, including wait-and-seek periods. This is repeated until the first self-heating is detected. This happens for the reference scenario (black line) at 108 °C and 7.8 h, marked by SH-II. Then, the ARC switches to the self-heating mode and starts to follow the cell temperature. For the reference scenario, self-heating at this stage is not sustained, and after a period of 1 h at 110 °C, an additional heating step is performed. At 119 °C and 10.1 h, marked by SH-III, another self-heating of the cell leads to switching to the exothermic mode of the ARC again. The temperature monotonically increases throughout the subsequent 13 h until the thermal runaway is triggered at 174 °C and 23 h marked by TR.

Fig. 2a shows the effect of variations in SEI composition at 90 vol% LEDC (dashed line) and 0 vol% (dash-dotted line). No significant difference between the scenarios can be observed during the preheating phase and in the first six heating steps. At 98 °C and 6.9 h, marked by SH-I, significant self-heating is observed for the 90 vol% LEDC-case, which is 11 °C earlier compared to the reference case. Temperature increases monotonically without a further heating step, and the cell reaches the thermal runaway at 173 °C and 24.5 h. For the inorganic SEI without LEDC, one additional heating step compared to the reference case is needed, with self-heating starting only at 127 °C and 10 h. This makes a difference of +9 °C compared to the reference case. Yet, the cell goes faster into the thermal runaway. It can be concluded that SEI composition has a notable effect on thermal safety of a battery, with cells with more LEDC, *e.g.* due to less ageing, being more likely to enter a thermal event, leading finally to thermal runaway.

In Fig. 2b, the effect of SEI thickness on thermal runaway is presented. The behaviour during the preheating and first heating steps shows no deviation from that of the reference case. The thin SEI enters one heating step earlier into the self-heating phase than the reference case. Self-heating progresses and causes a rapid thermal runaway at 174 °C and 23.3 h. In contrast, the thick SEI case reaches the self-heating only after a further heating step, at around 127 °C and 10 h, and enters the final runaway phase earlier, at 174 °C and 20.6 h. According to this analysis, SEI thickness also significantly impacts battery safety, with a thicker SEI leading less quickly to self-heating. It should be noted that the results presented here are specific to the investigated system of a graphite anode combined with EC/EMC 1.2 M $LiPF_6$ liquid electrolyte. For example, in the case of Li metal and all solid state batteries, a decrease in safety with increasing SEI thickness was found.[54,55]

Finally Fig. 2c shows the impact of H_2O impurities. As in the variations before, no difference is observed until 108 °C and 7.8 h. Then, the cell with high H_2O amounts enters one heating-step earlier into the sustained self-heating phase. The thermal runaway is reached at 174 °C and 27 h. In contrast, the curves for low and medium water amounts are identical. Thus, high amounts of H_2O seem to be detrimental to thermal safety of Li-ion batteries.

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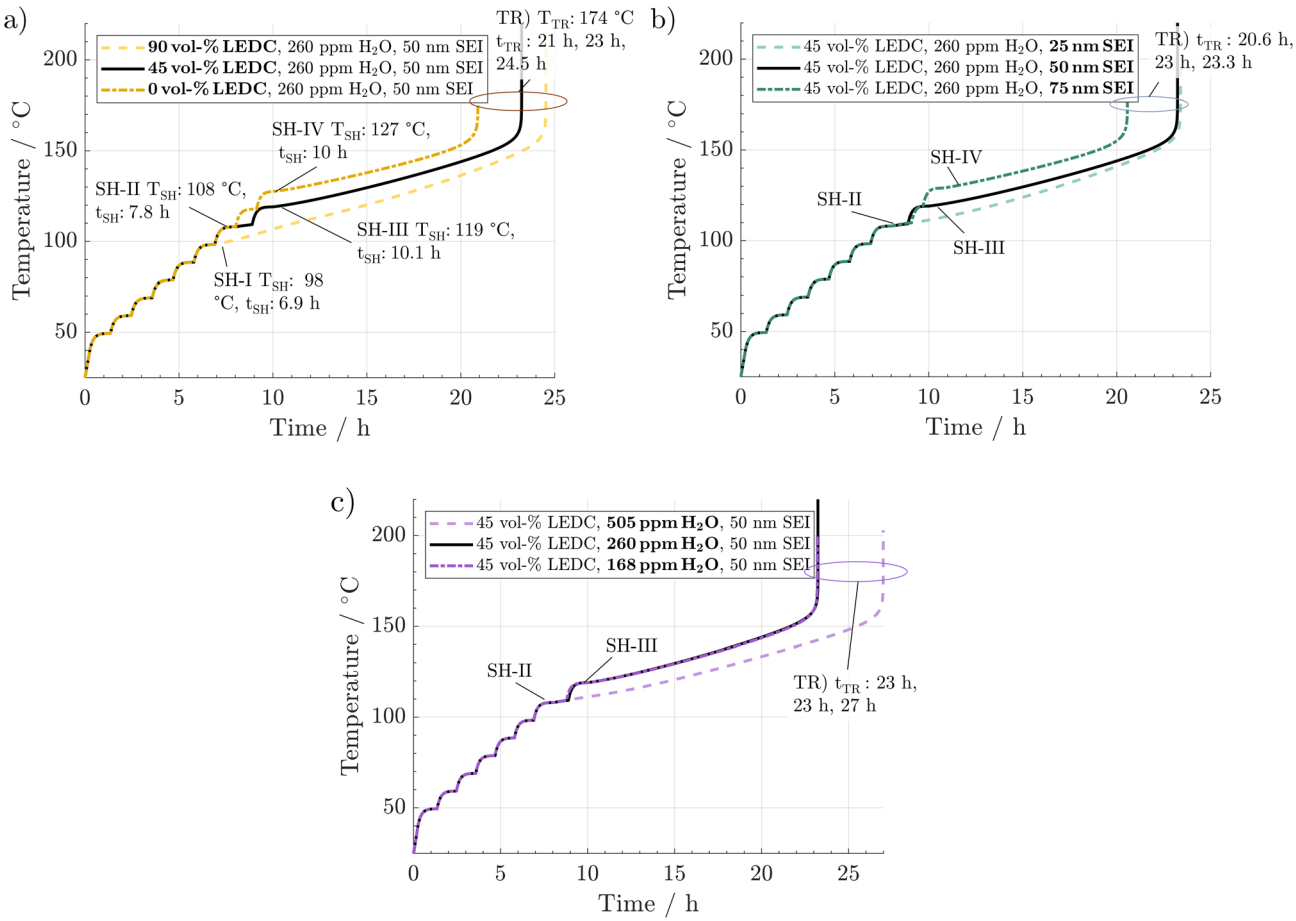

Fig. 2 ARC simulations with variations of (a) SEI composition, (b) SEI thickness, and (c) water content originating from the electrodes. The solid black line refers to the reference case in all subfigures. The lighter dashed line refers to the scenario featuring the lower self-heating temperature T_{SH}. The dash-dotted darker line indicates the scenario with the higher self-heating temperature. Characteristic self-heating temperatures, self-heating times, t_{SH}, and runaway times, t_{TR}, are indicated in the graphs.

From the above-described results, we conclude that all three parameters significantly impact thermal safety. The most apparent difference between all cases is the change in self-heating temperature and the time from there until the thermal runaway is reached. For a better comparison, these key parameters are summarised in Fig. 3. Here, a clear trend can be observed. A lower self-heating temperature corresponds to a longer time until the cell reaches the thermal runaway eventually. These opposing trends of earlier self-heating but later thermal runaway pose a fundamental question as to how to produce inherently safer batteries: while higher self-heating temperatures can be interpreted as safer, a shorter time until reaching the thermal runaway, and with this, a virtually unstoppable thermal event could be considered unsafe. Manufacturers must perform a risk analysis to make a good trade-off for these safety-critical parameters. While discussing the case study, including 27 variations, in Section 3.3, we offer an alternative metric to include both characteristics in the safety assessment of Li-ion batteries. Note that the temperature for the start of the rapid thermal runaway is identical in all cases. Similarly, as reported in the literature, this is almost exclusively caused by the onset of cathode active material decomposition. In our study, the cathode material was not varied. Thus, this behaviour was expected.

In the following section, a deeper analysis of processes during the thermal abuse tests is performed to reveal the origin of the manufacturing- and ageing-specific differences in safety behaviour.

3.2 Analysis of processes during thermal abuse

To understand the main reactions impacting the self-heating and their sensitivity to the SEI state and water impurities, we here analyse the progression of reactions, related species concentrations and produced or consumed heat during the thermal misuse. For better readability, the figures display only those reactions that are substantially influenced by SEI composition or water content. For the same reason, all SEI-forming reactions, namely OSP, ISP, and LSP, are summed up. For individual contributions, see the ESI.† Analysis starts at 40 °C, where the first degradation reactions are observed (Fig. 4): the initial decomposition of the conductive salt is triggered by a shift of the equilibrium towards the decomposition product with increasing temperature. Furthermore, the

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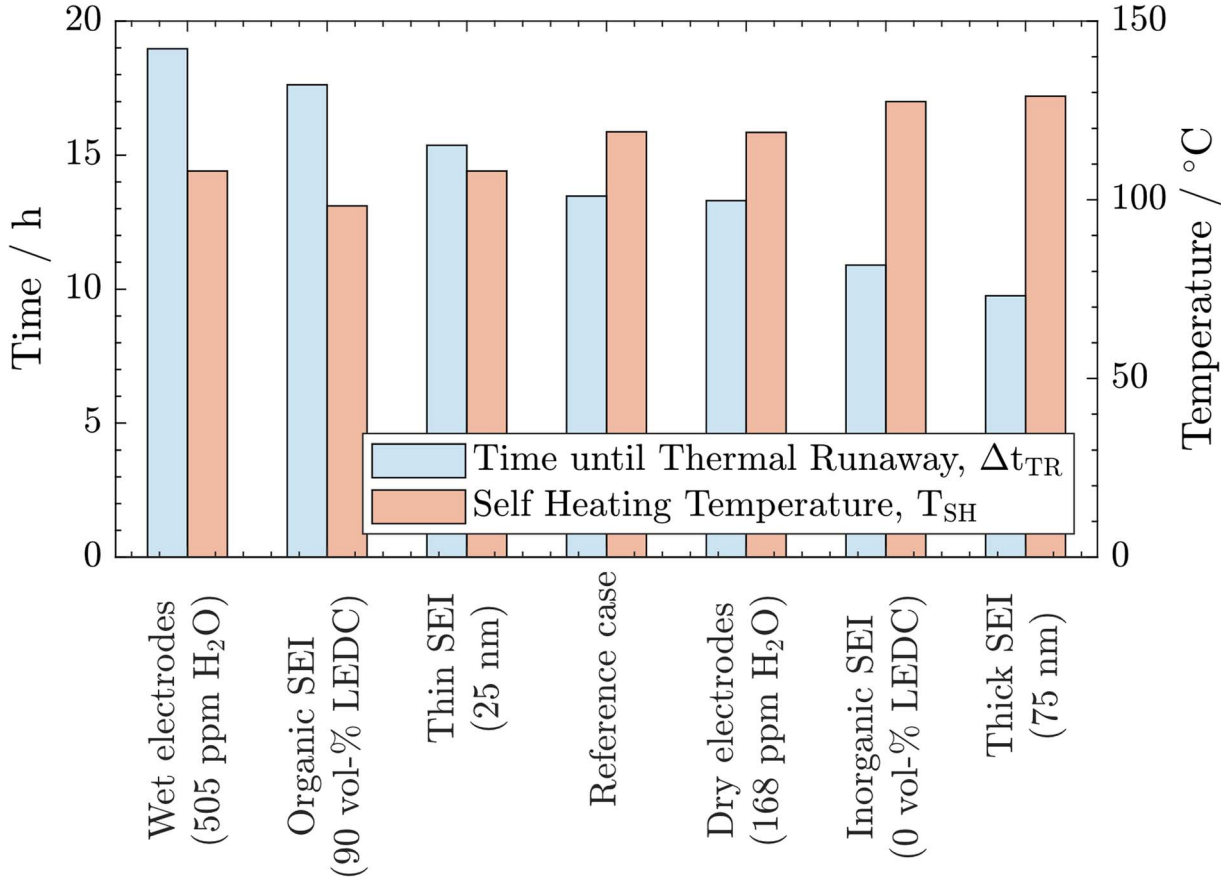

Fig. 3 Comparison of self-heating temperature T_{SH} and time until the thermal runaway is reached after the first exothermic phase Δt_{TR} as key indicators for all cases.

decomposition of PF_5 (PFD) and POF_3 (POFD) starts at 60 °C, peaks around 75 °C, and ends for the first time around 110 °C (SH-II). The HF released from these reactions initiates the decomposition of Li_2CO_3 (ISD), which, thus, happens simultaneously to PFD and POFD. Note that the concentration of Li_2CO_3 is only slightly declining due to the only small amounts of H_2O, and thus reactant HF, whereas it is produced in significant rates also from LEDC decomposition. Decomposition of PF_5, POF_3 and Li_2CO_3 happen simultaneously, and the ratio of produced heat by Li_2CO_3 decomposition *vs.* the consumed heat by PF_5 and POF_3 decomposition is always below 1 and decreases with temperature. Thus, the reactions caused by salt decomposition products, *i.e.* PFD, POFD, and ISD, act as heat sinks. The cause of the extinction of these reactions is the depletion of the necessary reactant H_2O. As the reaction rates and, therefore, their interplay are strongly dependent on water availability, we continue the discussion when analysing the wet and dry case scenarios. Reactions connected to salt decomposition are complemented by the exothermic decomposition of LEDC with notable reaction rates occurring above 80 °C. Eventually, by increased decomposition of LEDC and already decreasing endothermic heat due to H_2O depletion, the first self-heating starts at 108 °C (SH-II). This phase is, however, not self-sustaining because the exothermic LEDC decomposition slows down when much of the available LEDC has been consumed, so that a further heating step is required to trigger a thermal event. The transition to continuous self-heating of the battery at 118 °C (SH-III) is caused by the almost complete decomposition of $LiPF_6$, which reduces the endothermic heat to nearly 0. The self-heating phase until the rapid thermal runaway is dominated by the re-formation of the organic SEI, *i.e.* LEDC, and the inorganic SEI, *i.e.* Li_2CO_3, and further decomposition of LEDC. The declining concentration of LiOH denotes its decomposition to Li_2O (LSD) around 120 °C. The amount of LiOH introduced into the system by H_2O in the anode after drying is too small to produce notable amounts of exothermic heat. The thermal runaway is eventually set in motion by the decomposition of the cathode active material (CSD) starting around 150 °C. The thus produced O_2 triggers the subsequent solvent combustion, first of EC (ECD) and then of EMC (EMCD); see the ESI.† The combustion product H_2O, in turn, triggers an exponential increase in PF_5 (PFD) and POF_3 (POFD) decomposition reactions; despite being endothermic, they cannot compensate for the strongly exothermic reactions. Simultaneously, the exothermic decomposition of Li_2CO_3 is also re-initiated by HF produced from the PF_5 and POF_3 decomposition. Finally, it should be noted that above 130 °C, the LEDC concentration is kept at around zero, as any generated LEDC is